# Helium–CaF Collision Rates after a 4 K Buffer-Gas Source

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**Abstract**

This note gives an analytical estimate of the collision rate experienced by CaF molecules after they leave the aperture of the cryogenic source described by the user. It reports the local collision rate as a function of distance, the expected number of collisions in each successive 1 cm interval, and the cumulative collision number. The model is intended as an engineering estimate. In the first few aperture diameters the expansion is hydrodynamic, so a quantitatively reliable treatment requires a DSMC or equivalent rarefied-gas simulation.

## 1 Source parameters

The calculation uses:

| | |
|---|---|
| Helium flow | $2.0 \times 10^1$ sccm, continuous |
| Cell temperature | 4 K |
| Exit aperture | 4 mm diameter, $a = 2$ mm radius |
| CaF production | $1.0 \times 10^1$ ns ablation pulse at $1.0 \times 10^1$ Hz |
| CaF forward speed | $v_{\mathrm{CaF}} = 1.60 \times 10^2\,\mathrm{m\,s^{-1}}$ |
| Reference elastic cross section | $\sigma = 1 \times 10^{-18}\,\mathrm{m}^2$ |
| Pressure at a $3.00 \times 10^2$ K wall, $5.0 \times 10^1$ cm away | $2 \times 10^{-5}$ mbar |

The CaF production is pulsed, whereas helium is flowed continuously. Therefore, the quantities below apply to each CaF molecule in each pulse. The $1.0 \times 10^1$ Hz repetition rate does not enter the collision probability per molecule, provided the ablation pulse does not substantially disturb the stationary helium-flow field.

## 2 Analytical model

The helium particle flow corresponding to $2.0 \times 10^1$ sccm is

$$\dot{N}_{\mathrm{He}} = 8.96 \times 10^{18}\ \mathrm{s}^{-1}. \tag{1}$$

Outside the aperture, the on-axis helium density is approximated by a regularized hemispherical expansion,

$$n_{\mathrm{He}}(z) = \frac{\dot{N}_{\mathrm{He}}}{\pi u_{\mathrm{He}}\,(z^2 + a^2)}, \tag{2}$$

where $u_{\mathrm{He}} = 1.60 \times 10^2\,\mathrm{m\,s^{-1}}$ and $a = 2$ mm. The $a^2$ term prevents an unphysical divergence at the aperture. This regularization is a convenient interpolation, not a rigorous solution of the near-field gas dynamics.

The local CaF–He collision rate is

$$\Gamma(z) = n_{\mathrm{He}}(z)\,\sigma\,\langle v_{\mathrm{rel}}\rangle. \tag{3}$$

For co-propagating CaF and cold helium, we use

$$\langle v_{\rm rel}\rangle = 1.45 \times 10^2\,\mathrm{m\,s^{-1}}, \tag{4}$$

representative of the helium thermal speed at 4 K. Combining the preceding expressions gives

$$\boxed{\Gamma(z) = \frac{2.58}{z^2 + (0.002)^2}\ \mathrm{s}^{-1}}, \tag{5}$$

with $z$ expressed in metres and $\sigma = 1 \times 10^{-18}\,\mathrm{m}^2$.

The expected number of collisions while the molecule travels from $z_1$ to $z_2$ is

$$\Delta N(z_1, z_2) = \int_{z_1}^{z_2} \frac{\Gamma(z)}{v_{\rm CaF}}\,\mathrm{d}z \tag{6}$$

$$= \frac{K}{a v_{\rm CaF}}\left[\tan^{-1}\!\left(\frac{z_2}{a}\right) - \tan^{-1}\!\left(\frac{z_1}{a}\right)\right], \tag{7}$$

where $K = 2.58\,\mathrm{m^2\,s^{-1}}$. The cumulative collision number from the aperture to distance $z$ is therefore

$$\boxed{N(0, z) = \frac{K}{a v_{\rm CaF}} \tan^{-1}\!\left(\frac{z}{a}\right)}. \tag{8}$$

All rates and collision counts scale linearly with the assumed cross section:

$$\Gamma,\ \Delta N,\ N \propto \frac{\sigma}{1 \times 10^{-18}\,\mathrm{m}^2}. \tag{9}$$

# 3 Distance-resolved results

The column "collisions in this centimetre" is the expectation value accumulated while the molecule travels from $(z-1)$ to $z$ centimetres. "Cumulative from aperture" includes the model-dependent first centimetre. Since the first few millimetres are hydrodynamic, the final column also gives the more robust cumulative count starting at 1 cm.

Table 1: Local and cumulative CaF–He collision estimates for $\sigma = 1 \times 10^{-18}\,\mathrm{m}^2$.

| $z$ (cm) | $\Gamma(z)$ ($\mathrm{s}^{-1}$) | mean time between collisions | collisions in this cm | cumulative from aperture | cumulative from 1 cm |
|---|---|---|---|---|---|
| 1 | 2.48e+04 | 40.3 $\mu$s | 11.073 | 11.073 | 0.000 |
| 2 | 6.39e+03 | 156.6 $\mu$s | 0.788 | 11.861 | 0.788 |
| 3 | 2.85e+03 | 350.4 $\mu$s | 0.267 | 12.128 | 1.055 |
| 4 | 1.61e+03 | 621.7 $\mu$s | 0.134 | 12.262 | 1.189 |
| 5 | 1.03e+03 | 970.5 $\mu$s | 0.080 | 12.342 | 1.269 |
| 6 | 716 | 1.40 ms | 0.054 | 12.396 | 1.323 |
| 7 | 526 | 1.90 ms | 0.038 | 12.434 | 1.361 |
| 8 | 403 | 2.48 ms | 0.029 | 12.463 | 1.390 |
| 9 | 318 | 3.14 ms | 0.022 | 12.485 | 1.412 |
| 10 | 258 | 3.88 ms | 0.018 | 12.503 | 1.430 |
| 11 | 213 | 4.69 ms | 0.015 | 12.518 | 1.445 |
| 12 | 179 | 5.58 ms | 0.012 | 12.530 | 1.457 |
| 13 | 153 | 6.55 ms | 0.010 | 12.541 | 1.467 |
| 14 | 132 | 7.60 ms | 0.009 | 12.549 | 1.476 |
| 15 | 115 | 8.72 ms | 0.008 | 12.557 | 1.484 |
| 16 | 101 | 9.92 ms | 0.007 | 12.564 | 1.491 |
| 17 | 89.3 | 11.20 ms | 0.006 | 12.570 | 1.497 |
| 18 | 79.6 | 12.56 ms | 0.005 | 12.575 | 1.502 |

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| $z$ (cm) | $\Gamma(z)$ ($s^{-1}$) | mean time between collisions | collisions in this cm | cumulative from aperture | cumulative from 1 cm |
|---|---|---|---|---|---|
| 19 | 71.5 | 13.99 ms | 0.005 | 12.580 | 1.507 |
| 20 | 64.5 | 15.51 ms | 0.004 | 12.584 | 1.511 |
| 21 | 58.5 | 17.09 ms | 0.004 | 12.588 | 1.515 |
| 22 | 53.3 | 18.76 ms | 0.003 | 12.591 | 1.518 |
| 23 | 48.8 | 20.51 ms | 0.003 | 12.594 | 1.521 |
| 24 | 44.8 | 22.33 ms | 0.003 | 12.597 | 1.524 |
| 25 | 41.3 | 24.23 ms | 0.003 | 12.600 | 1.527 |
| 26 | 38.2 | 26.20 ms | 0.002 | 12.603 | 1.529 |
| 27 | 35.4 | 28.26 ms | 0.002 | 12.605 | 1.532 |
| 28 | 32.9 | 30.39 ms | 0.002 | 12.607 | 1.534 |
| 29 | 30.7 | 32.60 ms | 0.002 | 12.609 | 1.536 |
| 30 | 28.7 | 34.89 ms | 0.002 | 12.611 | 1.538 |
| 31 | 26.8 | 37.25 ms | 0.002 | 12.613 | 1.539 |
| 32 | 25.2 | 39.69 ms | 0.002 | 12.614 | 1.541 |
| 33 | 23.7 | 42.21 ms | 0.002 | 12.616 | 1.543 |
| 34 | 22.3 | 44.81 ms | 0.001 | 12.617 | 1.544 |
| 35 | 21.1 | 47.48 ms | 0.001 | 12.618 | 1.545 |
| 36 | 19.9 | 50.23 ms | 0.001 | 12.620 | 1.547 |
| 37 | 18.8 | 53.06 ms | 0.001 | 12.621 | 1.548 |
| 38 | 17.9 | 55.97 ms | 0.001 | 12.622 | 1.549 |
| 39 | 17.0 | 58.96 ms | 0.001 | 12.623 | 1.550 |
| 40 | 16.1 | 62.02 ms | 0.001 | 12.624 | 1.551 |
| 41 | 15.3 | 65.16 ms | 0.001 | 12.625 | 1.552 |
| 42 | 14.6 | 68.37 ms | 0.001 | 12.626 | 1.553 |
| 43 | 14.0 | 71.67 ms | 0.001 | 12.627 | 1.554 |
| 44 | 13.3 | 75.04 ms | 0.001 | 12.628 | 1.555 |
| 45 | 12.7 | 78.49 ms | 0.001 | 12.629 | 1.556 |
| 46 | 12.2 | 82.02 ms | 0.001 | 12.629 | 1.556 |
| 47 | 11.7 | 85.62 ms | 0.001 | 12.630 | 1.557 |
| 48 | 11.2 | 89.30 ms | 0.001 | 12.631 | 1.558 |
| 49 | 10.7 | 93.06 ms | 0.001 | 12.632 | 1.559 |
| 50 | 10.3 | 96.90 ms | 0.001 | 12.632 | 1.559 |

## 4 Interpretation and limitations

The model predicts that the collision rate decreases approximately as $z^{-2}$ once $z \gg a$. It also predicts that most post-aperture collisions occur very near the nozzle. That qualitative conclusion is robust, but the exact cumulative collision count from $z = 0$ is not: the first few millimetres are precisely where the hemispherical free-expansion approximation is least valid.

For this reason, the column "cumulative from 1 cm" is often the safer quantity to use when estimating additional collisions after the strongly collisional near-nozzle region. A source-specific DSMC calculation would be needed to replace the regularized analytical model in that region.

The pressure measured $5.0 \times 10^1$ cm away at a room-temperature wall characterizes the diffuse chamber background rather than the cold directed jet. It should therefore be treated separately from the on-axis expansion model above.

## 5 Reference

The source geometry and operating context were supplied by the user and refer to the Optics Express article with DOI 10.1364/OE.560951.